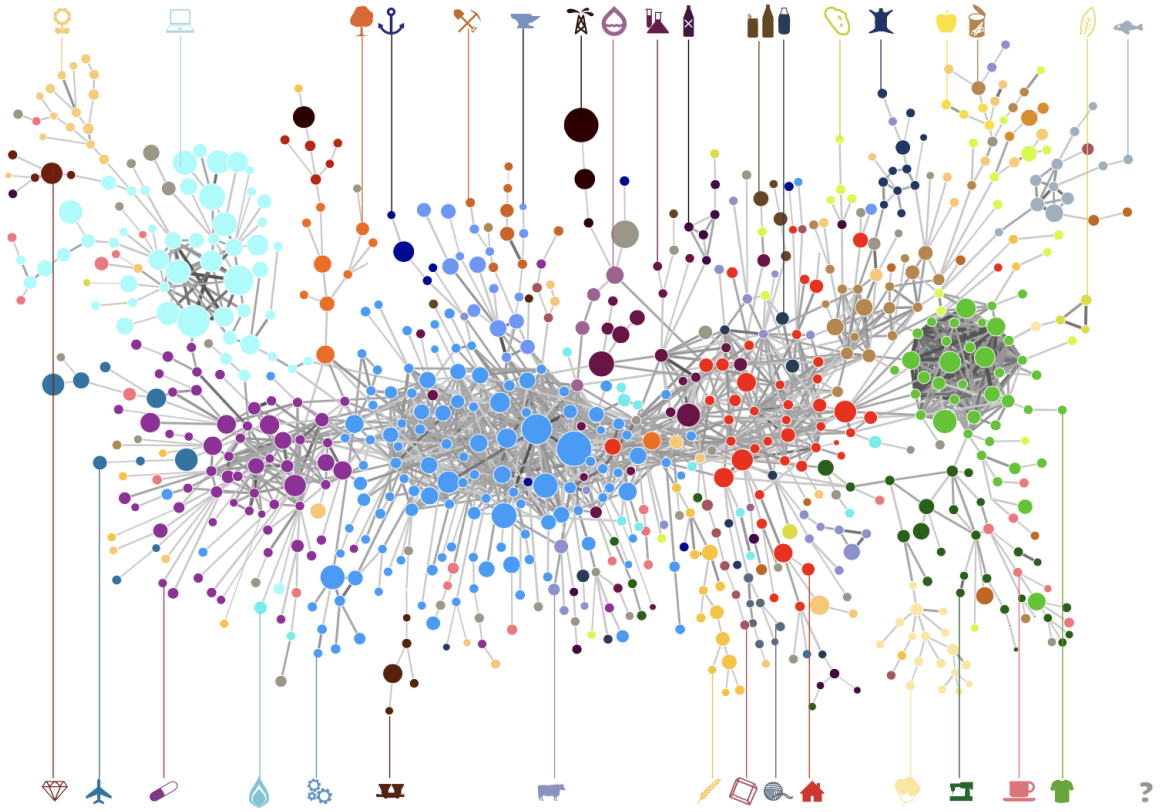


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# Optimal Industry Development Strategy Using Economic and Product Complexity

# Thailand

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# Optimal Industry Development Strategy for Thailand

## 0. Executive Summary

In this report, we aim to uncover the economic path Thailand went through during the period from 2013 to 2023. Specifically, we dive into the topic through a macroeconomic overview of Thailand from the perspective of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect countries' production capability and production structure, respectively. We find that during the period of our interest, Thailand demonstrated a low level of economic growth with an average annual GDP growth rate of -1.0% between 2019 and 2023, with a decreased economic complexity compared to peer economies.

The mixed progress during the period, we believe, can be partially explained through the analysis of the economic and product complexity structure of Thailand on the industry group level, as well as the stagnant macro and microeconomic strategies. The product classification system we adopt allows us to measure the realized industry development for various industry groups and to observe that Thailand regressed in developing most of its industries during the period while its developing counterparts and economies around the globe also adopted complex and progressive paths with variable degrees of success.

Based on our analysis of the opportunity gain and sensitivity analysis of major industry groups, Thailand could further its gains by prioritizing Energy Drilling and Mining and Clothing and Accessories. We further note that Thailand has a heavily mismatched development strategy that indicates a low level (compared to other developing countries) of structural optimality on the complexity level, having an adjusted R-square of around -5.8% in 2018 and a much higher score of 17.9% in 2023, demonstrating shift and improvement in development. Since a significant positive correlation exists between GDP growth and the structural optimality index of an economy as we define and calculate, Thailand indeed has a suboptimal growth experience as compared with fast-growing economies both on a regional scale and on a global scale and has the potential to further its economic gains through industrial structure optimization as suggested in the paper.

The five years between 2019 and 2023 showed a significant worsening in Thailand's economic growth, lowering its position on the global scale. Its GDP stayed stagnant below the pre-pandemic levels, and it saw a significant decrease in the ECI, particularly compared to its peer countries. Its current strategy still has the viable potential upside to increase its performance and optimize its structure from its current status.

# 1. Report Overview

In this report, we aim to uncover the economic path Thailand went through during the period from 2019 to 2023, before and after the COVID pandemic, compared to its performance from 2013 to 2018, before and after the military coup of 2014. Specifically, we dive into the topic through a macroeconomic overview of Thailand from the perspective of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect countries' production capability and production structure, respectively. We then analyze the industrial framework and changes of Thailand during the period based on quantitative measures such as the Product Complexity Index (PCI) and Product Family Complexity Index (PfCI), followed by a detailed analysis of how such changes have matched with development strategies as quantitatively measured by opportunity gains on an industry-level. We conclude our report with macro-level results and predictions of Thailand, as deduced from regression models we have established on a global scale, which take into account countries' stances in their product space.

## 2. Macroeconomic Results

### 2.1 GDP Ranking and Changes

Figure 1a illustrates the time-series GDP results for Thailand and other major Asian economies (Indonesia, Singapore, China, and Japan) from 2013 to 2018. Similar to the other countries, Thailand displayed an eventual upward trend over the period; however, its growth was more consistent than Japan's growth and less consistent than China's growth. Thailand's average annual GDP growth rate (4.2%) is in the middle, as it is greater than Japan's rate (1.16%) but less than China's rate (7.1%). Noteworthy is that Thailand's average annual GDP growth rate is almost at the same level as the annual growth rate of the other four leading asian economies, which have a geometric average growth rate of 3.0% and an arithmetic average growth rates of 3.2%.

Figure 1b shows the GDP levels for Thailand and two of its peer countries (Malaysia and Chile) from 2019 to 2023. Similar to the other countries, Thailand displayed a drop in GDP from 2019 to 2020 because of the pandemic, with all GDPs rebounding in 2021. Over these five years, Thailand's average annual GDP growth rate (-1.0%) is the lowest of the three in the cohort, as it is less than Malaysia's rate (2.08%) and also less than Chile's (4.55%). It is notable that the GDP levels in Thailand stayed relatively stagnant compared to Malaysia's and Chile's, which both closed the gap between their GDP and Thailand's.

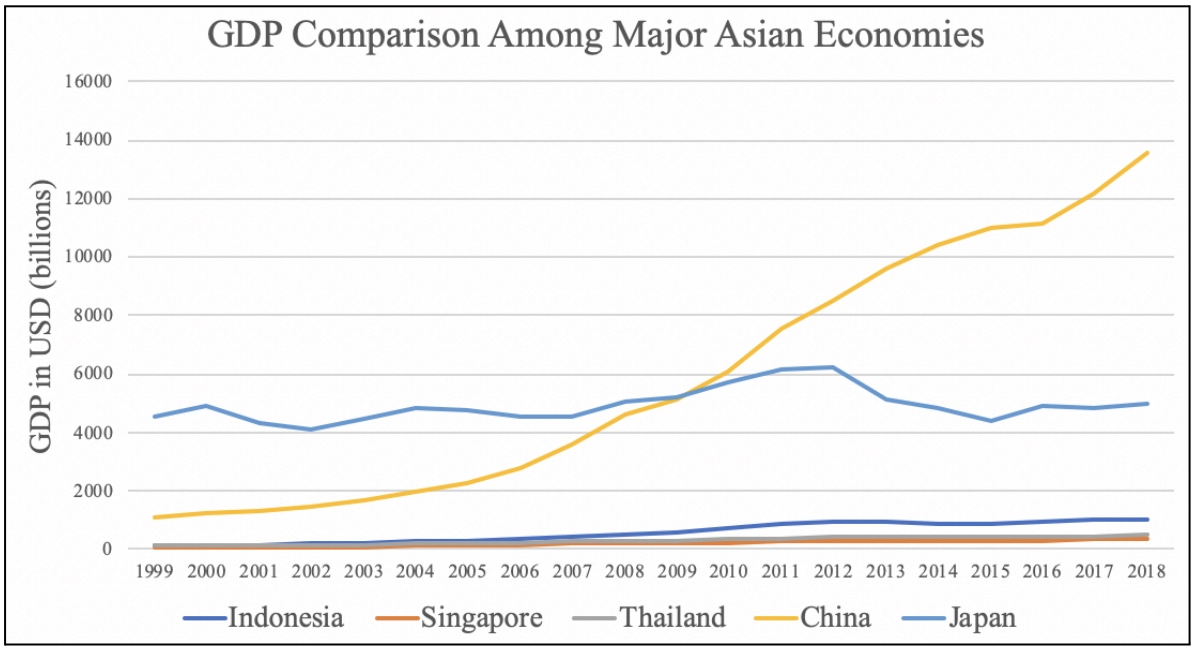



Figure 1a. GDP Comparison Among Asian Economies (2013-2018)

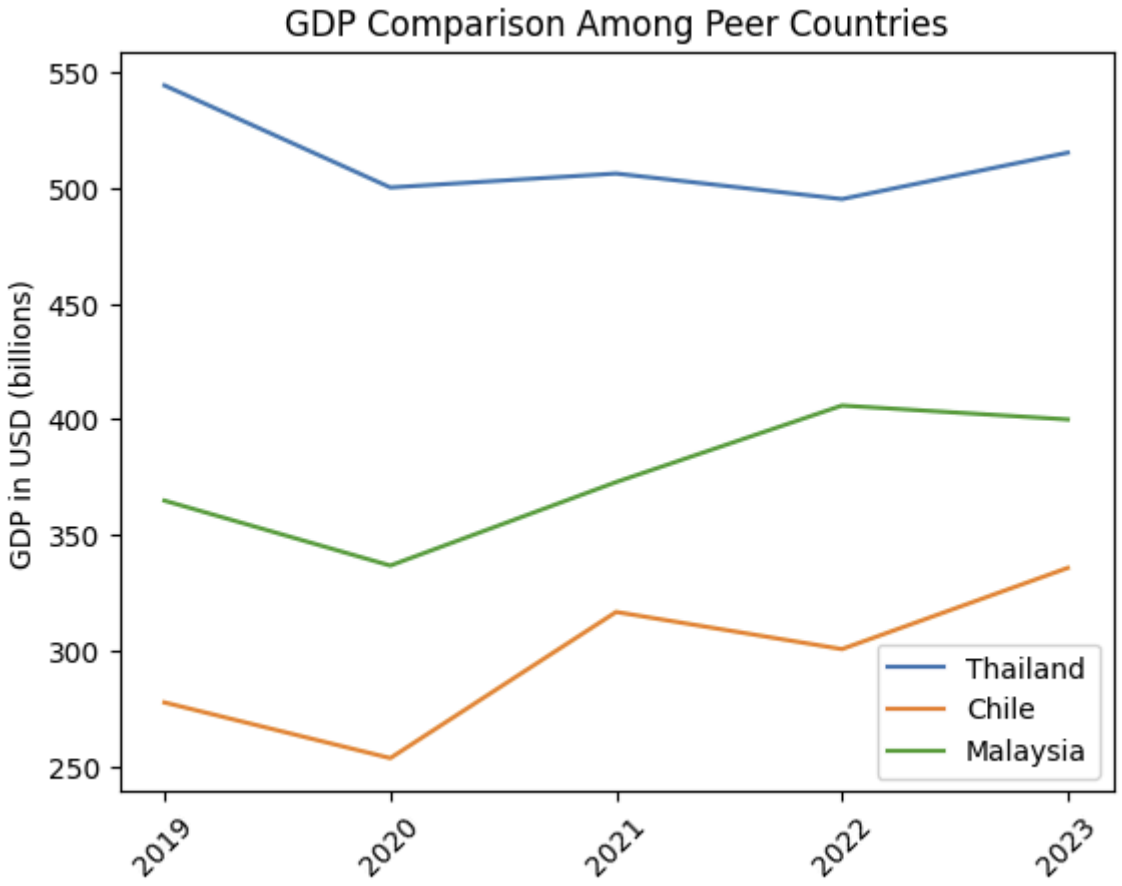



Figure 1b. GDP Comparison Among Peer Countries (2019-2023)

**2.2 ECI Ranking and Changes**

The Economic Complexity Index (ECI), as defined by Ricardo Hausmann and Cesar Hidalgo in 2009 to explain the knowledge accumulated in a country's population, is illustrated in Figures 2a and 2b.

As shown, Thailand exhibits a slight decrease in production capacity from 2013 to 2018. Compared to its Asian peers Indonesia, China, Japan, and Singapore, Thailand's ECI level similarly stays relatively constant but mirrors the overall trend of gradual decrease from 2013 to 2018. Thailand's ECI level is in the middle compared to these leading Asian economies during this period, with its ECI level consistently below that of Singapore, Japan, and China but consistently above that of Indonesia.

However, Thailand exhibits a large decrease in production capability from 2019 to 2023. Thailand is in a weak position in ECI terms, as its ECI change of approximately -450% (to standardize for a negative starting ECI value in 2019) is less than both India's and Chile's, which are around 200% (again, to standardize for negative starting ECI values in 2019).

The comparable countries maintain a large change in ECI level between 2019 and 2020, with a markedly smaller change between 2020 and 2021. Thailand exhibits a modest increase in production capability in 2020 to levels not seen prior, with a pandemic-driven demand for higher-tech products being one of the reasons.

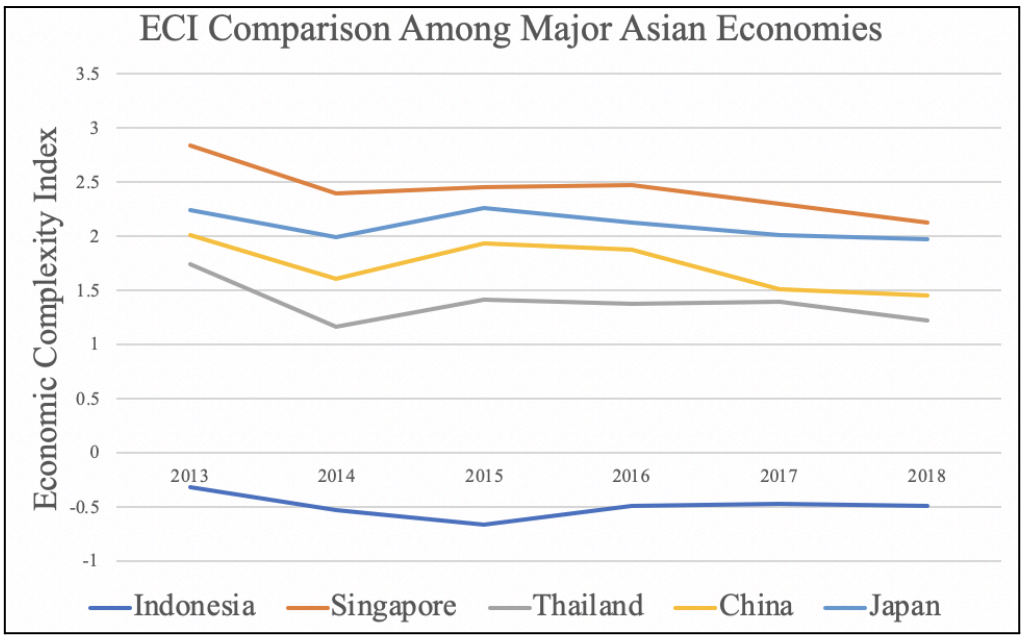



Figure 2b. ECI Comparison Among Major Asian Economies (2013-2018)

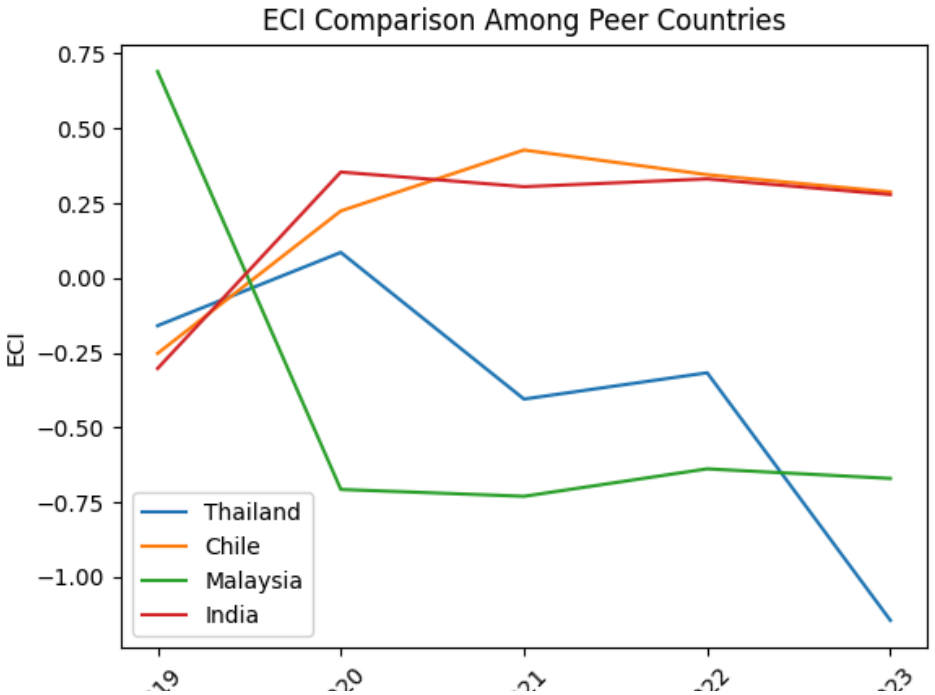



Figure 2b. ECI Comparison Among Peer Countries (2019-2023)

Figure 3 offers a further comparison of GDP growth and ECI over the period from 2013 to 2023 for Thailand. Similar to the modest GDP growth from 2013 to 2018 and the corresponding slight decline in ECI, the mild GDP decrease during the period from 2019 to 2023 is accompanied by a volatile decrease in ECI. The results show that the economic growth experienced by Thailand during the period from 2013 to 2018 is only partially consistent with the predictions by ECI, and that Thailand made weak progress towards effectively developing a competitive product space that helps to accumulate complicated knowledge and technology. For the period 2019 to 2023, the ECI followed GDP trends,

which can be interpreted as regressions in production. In order to provide a more concrete answer to why this has happened, we will dive into an analysis of Thailand's production structure and its product space designs in the next few sections.

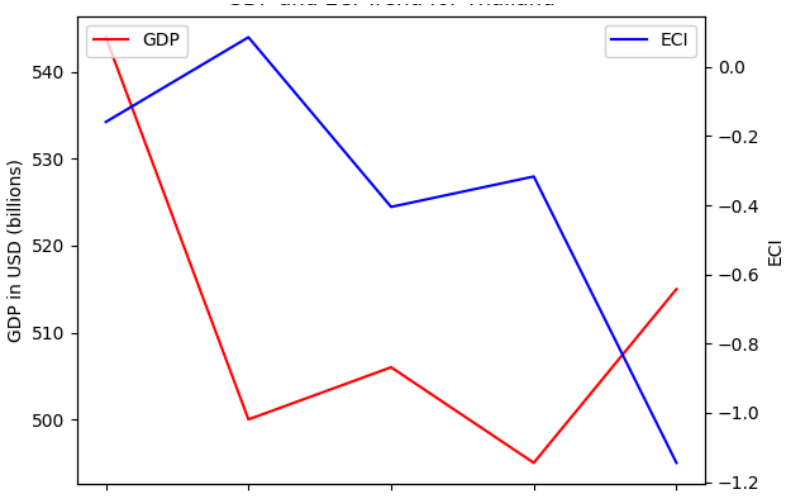



Figure 3. GDP Growth and ECI for Thailand (2019-2023)

# 3. Production Analysis

## 3.1 Product Categorization

Our main data source in this research is the UN Comtrade Website (https://comtrade.un.org), which provides us with valuable information on countries' export values on an annual basis, including highly specific export data for more than 5000 categories of products. In order to have a holistic vision of the product space of a country, however, a level of categorization is required. Products in our data source are labeled from 010000-999999, with the first two digits representing a general category to which the product belongs. Based on the 99 categories extracted from the label, we further group the products into the 18 most widely-accepted industries, as summarized in Table 1.

Note that the High-complexity Products industry group is not a distinct industry but rather an aggregated industry that encompasses the most complicated products from the most technologically-advanced industries (as measured by PCI). This industry serves as an indicator for us to quantitatively evaluate how well a country has performed in developing cutting-edge technologies and closing its technological gap with advanced economies, which has become a topic of wide concern today. Indeed, when we rank the 18 categories based on PfCI (an equivalent concept of PCI as calculated based on the 18 categorized product groups), the High-complexity Products industry has the highest PfCI among all industry groups.

| Category | Description |
|---|---|
| **Animals and By-products** | All kinds of meat, livestock and related by-products |
| **Vegetables, Fruits and Plants** | All kinds of plants, edible vegetables and fruits |
| **Food and Preparations** | Edible preparations, raw cooking materials and appliances |
| **Clothing and Accessories** | All kinds of fabrics, filament, clothing and accessories |
| **Textiles** | All kinds of textile articles and related products |
| **Wood Products** | Book, paper and other elementary products related to wood |
| **Metals** | Mining of all sorts of raw Metals |
| **Refined and Processed Stones** | Refined or processed stone products such as pearl and ceramics |
| **Energy Drilling and Mining** | Mining industry for energy, minerals, stone materials, etc. |
| **Daily Instruments** | Basic daily instruments such as sport requisites, timing and measuring tools |
| **Raw Manufacturing** | Basic products made from plastic, rubber, leather, etc. |
| **Chemical Manufacturing** | Advanced chemical products and pharmaceutical products |
| **Chemicals** | Raw chemical materials and by-products |
| **Machinery and Construction** | Machinery and mechanical appliances, including electrical equipment |
| **Transportation** | Vehicles and locomotives including aircraft, ship, railway, etc. |
| **Weapons** | Explosives, arms and ammunition |
| **Services and Utilities** | Financial, business, legal, communication, travel Services and Utilities, etc. |
| **High-complexity Products** | Agglomeration of advanced products with highest PCIs |

Table 1. Categorization of Products

We further observe a positive correlation between the GDP and the export of High-complexity Products. A higher adjusted R-square exists between GDP and High-complexity Products exports, indicating that more successful economies, as measured by GDP, tend to be those that are capable of producing and exporting High-complexity Products as we define them. Tables 2a and 2b show the OLS results in a universe of 107 and 21 typical countries, respectively.

| | High-complexity Products | Food and Preparations | Animals and By-products |
|---|---|---|---|
| Adj. R-squared | 0.075 | 0.006 | 0.002 |
| F-statistic | 9.556 | 1.638 | 1.175 |
| No. Observations | 107 | 107 | 107 |

Table 2a. OLS Regression Results of GDP against Percentage Volume of Exports

| | High-complexity Products | Food and Preparations | Animals and By-products |
|---|---|---|---|
| Adj. R-squared | 0.149 | 0.003 | -0.012 |
| F-statistic | 4.510 | 1.059 | 0.7542 |
| No. Observations | 21 | 21 | 21 |

Table 2b. OLS Regression Results of GDP against Volume of Percentage Exports

### 3.2 Realized Industry Development

Based on the aforementioned product categorization, we aim to capture Thailand's realized industry production strategy between 2013 and 2023 on a complexity level. In Hausmann and Hidalgo (2009), distance is defined to be a measure of how far a country is from a certain product, not in its product space. We extend this concept to all product groups and quantify how technically equipped a country is to develop the corresponding product groups. In other words, the shorter the distance between a country and a product group, the more capable a country is of developing products within that group. Defining this net distance change as the Realized Industry Development (RID), our export data allow us to calculate and compare the RID of Thailand and the 18 product groups in 2023 and 2018 through the difference, where a positive RID indicates a shortening of the distance (i.e., advancement of the industry), and a negative RID indicates the reverse. It is notable that the number of industries with positive RID has decreased significantly from 11 to 0 from 2018 to 2023.

| Rank | Category | RID | Rank | Category | RID |
|---|---|---|---|---|---|
| 1 | Chemicals | -0.035 | 10 | Vegetables, Fruits and Plants | 0.012 |
| 2 | Food and Preparations | -0.020 | 11 | Transportation | 0.013 |
| 3 | Energy Drilling and Mining | -0.013 | 12 | High-complexity Products | 0.018 |
| 4 | Clothing and Accessories | -0.005 | 13 | Wood Products | 0.022 |
| 5 | Metals | -0.003 | 14 | Precious Stones | 0.023 |
| 6 | Chemical Manufacturing | -0.002 | 15 | Machinery and Construction | 0.026 |
| 7 | Raw Manufacturing | -0.002 | 16 | Daily Instruments | 0.036 |
| 8 | Textile | 0.006 | 17 | Services and Utilities | 0.044 |
| 9 | Animals and By-products | 0.011 | 18 | Weapons | 0.057 |

Table 3a. Realized Industry Development (RID) of Thailand (2018)

| Category | RID |
|---|---|
| Animals and By-products | -0.436124 |
| Clothing and Accessories | -0.366881 |
| Precious Stones | -0.355473 |
| Energy Drilling and Mining | -0.317546 |
| Raw Manufacturing | -0.301390 |
| Wood Products | -0.286393 |
| Weapons | -0.285580 |
| Machinery and Construction | -0.285442 |
| Metals | -0.277343 |
| Services and Utilities | -0.272670 |
| Chemicals | -0.263896 |
| Textile | -0.260902 |
| Chemical Manufacturing | -0.242370 |
| Vegetables, Fruits and Plants | -0.219221 |
| Food and Preparations | -0.216025 |
| Daily Instruments | -0.199368 |
| Transportation | -0.128839 |
| High-complexity Products | -0.108764 |

Table 3b. Realized Industry Development (RID) of Thailand (2023)

### 3.3 Regional and Global Comparison

Table 4a below summarizes how the production strategy adopted by Thailand compares with selected leading economies in the world across different years, as measured in RID. Leading economies such as France (and Monaco), Greece, and Portugal gave a considerable level of attention to more advanced product groups such as Services and Utilities, Clothing, and Manufacturing in 2018 and Services and Utilities, Transportation and Chemical Manufacturing in 2023.

| Ranking | France, Monaco | Greece | Portugal |
|---|---|---|---|
| 1 | Services and Utilities | Vegetables, Fruits and Plants | Vegetables, Fruits and Plants |
| 2 | Chemical Manufacturing | Animals and By-products | Textile |
| 3 | Raw Manufacturing | Weapons | Animals and By-products |
| 4 | Transportation | Precious Stones | Wood Products |
| 5 | Energy Drilling and Mining | Transportation | Weapons |

Table 4a. RID Comparison across Leading Economies (2023)

Table 4b further examines Thailand's development strategy against those of emerging economies on a global scale. Reasonably, emerging economies such as China and Singapore

have devoted the majority of their efforts to intermediate industry groups, showing moderate shifts in strategies over the course of five years.

| Ranking | China | Singapore |
|---|---|---|
| 1 | Services and Utilities | High-complexity Products |
| 2 | Machinery and Construction | Transportation |
| 3 | Weapons | Raw Manufacturing |
| 4 | Chemicals | Machinery and Construction |
| 5 | Animals and By-products | Chemical Manufacturing |

Table 4b. RID Comparison across Emerging Economies (2023)

# 4. Product Space Design

## 4.1 Suggested Development Vector

As outlined above, distance gives us an idea of how far each new product is from a country's current mix of exports. As defined in Hausmann and Hidalgo (2009), distance is "the weighted proportion of products connected to goods p that country c is not exporting, i.e., MCP = 0." Based on this definition, the distance between a country and any product that the country is currently exporting (i.e., the country is a significant exporter of the product as compared to the global average) is zero. Similarly, opportunity gain quantifies the contribution of a new product in terms of opening up the doors to more and more complex products. Intuitively, a product that a country is currently exporting (i.e., MCP = 1) has an opportunity gain of zero. An unreasonable investment in a certain product p would lead to a relative decrement in the export of more profitable products and therefore lead to a negative opportunity gain of product p. The opportunity gain is not the only estimator of a country's level of economic development. However, among countries with comparable economic strength, those having higher ECI, or in other words, more reasonable economic structures are often in better shape.

| | Thailand | Chile | Malaysia | India |
|---|---|---|---|---|
| 0 | Energy Drilling and Mining | Services and Utilities | Services and Utilities | Food and Preparations |
| 1 | Clothing and Accessories | High-complexity Products | Chemical Manufacturing | Wood Products |
| 2 | Wood Products | Precious Stones | High-complexity Products | Weapons |
| 3 | Weapons | Transportation | Precious Stones | Raw Manufacturing |
| 4 | Chemical Manufacturing | Daily Instruments | Chemicals | Services and Utilities |
| 5 | Metals | Machinery and Construction | Vegetables, Fruits and Plants | Transportation |
| 6 | Chemicals | Chemical Manufacturing | Animals and By-products | Machinery and Construction |
| 7 | Machinery and Construction | Weapons | Machinery and Construction | Daily Instruments |
| 8 | Daily Instruments | Clothing and Accessories | Wood Products | High-complexity Products |
| 9 | Animals and By-products | Textile | Textile | Animals and By-products |
| 10 | Vegetables, Fruits and Plants | Raw Manufacturing | Food and Preparations | Vegetables, Fruits and Plants |
| 11 | Food and Preparations | Animals and By-products | Energy Drilling and Mining | Energy Drilling and Mining |
| 12 | Raw Manufacturing | Vegetables, Fruits and Plants | Raw Manufacturing | Chemicals |
| 13 | Textile | Food and Preparations | Metals | Chemical Manufacturing |
| 14 | Transportation | Energy Drilling and Mining | Transportation | Clothing and Accessories |
| 15 | Precious Stones | Chemicals | Weapons | Textile |
| 16 | High-complexity Products | Wood Products | Daily Instruments | Metals |
| 17 | Services and Utilities | Metals | Clothing and Accessories | Precious Stones |

Table 5. Opportunity Gain Ranking Between Comparable Countries (2023)

We define the ranking of opportunity gain for a country c, as the Suggested Development Vector (SDV) of c. Table 5 summarizes how the SDV of Thailand compares with a selected group of comparable economies in the world, with the industry groups ranked in decreasing order of significance.

Dark blue indicates positive opportunity gains, light blue indicates zero opportunity gains, and white indicates negative opportunity gains. The result shows that leading countries barely have any industries with positive opportunity gain. In other words, those countries are already in the most stable and efficient economic structure. Without the necessity of economic

transition, those countries merely need to maintain a steady and balanced economic development. In contrast, developing countries have nearly half of industries with positive opportunity gain and share common interests in Services and Utilities and Machinery and Construction. It can be concluded that Thailand shares the characteristics of developing countries, with a balanced economic structure that leans towards middle-tier complexity industries.

| Ranking | Japan | China | Germany | Republic of Korea | Thailand |
|---|---|---|---|---|---|
| 1 | Chemicals | Chemicals | Chemical Manufacturing | Services and Utilities | Energy Drilling and Mining |
| 2 | Metals | Raw Manufacturing | Machinery and Construction | Chemicals | Vegetables, Fruits and Plants |
| 3 | Machinery and Construction | Clothing and Accessories | Transportation | Raw Manufacturing | Animals and By-products |
| 4 | Transportation | Textile | Daily Instruments | Textile | Food and Preparations |
| 5 | Weapons | Machinery and Construction | High-complexity Products | Metals | Raw Manufacturing |
| 6 | Daily Instruments | Transportation | Services and Utilities | Transportation | Textile |
| 7 | High-complexity Products | Weapons | Chemicals | Daily Instruments | Machinery and Construction |
| 8 | Services and Utilities | Daily Instruments | Textile | High-complexity Products | Transportation |
| 9 | Textile | High-complexity Products | Weapons | Weapons | Precious Stones |
| 10 | Precious Stones | Precious Stones | Clothing and Accessories | Clothing and Accessories | High-complexity Products |
| 11 | Clothing and Accessories | Services and Utilities | Precious Stones | Precious Stones | Clothing and Accessories |
| 12 | Energy Drilling and Mining | Energy Drilling and Mining | Raw Manufacturing | Energy Drilling and Mining | Wood Products |
| 13 | Animals and By-products | Wood Products | Wood Products | Chemical Manufacturing | Chemicals |
| 14 | Wood Products | Animals and By-products | Energy Drilling and Mining | Wood Products | Services and Utilities |
| 15 | Chemical Manufacturing | Metals | Animals and By-products | Animals and By-products | Weapons |
| 16 | Food and Preparations | Chemical Manufacturing | Metals | Food and Preparations | Metals |
| 17 | Vegetables, Fruits and Plants | Vegetables, Fruits and Plants | Food and Preparations | Vegetables, Fruits and Plants | Chemical Manufacturing |
| 18 | Raw Manufacturing | Food and Preparations | Vegetables, Fruits and Plants | Machinery and Construction | Daily Instruments |
| ECI | 1.979430054 | 1.700687855 | 2.959929484 | 1.648971597 | 1.272660382 |

Table 6. SDV of Thailand and Other Sample Countries (2018)

### 4.2 Budgeted Industry Development Strategy and Interpretation

The optimal budgeted industry development strategy aims to solve the most efficient allocations under the budgeted constraint of different situations of heuristic total export growth. In order to obtain such an allocation portfolio, we formulate an optimization problem as the following:

$$\max_{x \in \mathbf{R}^{|P|}} \quad \left(\frac{k(x) - < k(x) >}{stdev(k(x))}\right)_C$$

Subject to

$$\sum_{p \in P} x_p \leq B$$

$$x \succcurlyeq 0$$

where k(x) is the second largest eigenvector of $M_{CC'}$ after adding $x$ to the original export vector, $C$ is the country that we are interested in, $P$ is the product space, $B$ is the heuristic budget.

If we can solve this problem, then we can know what the largest ECI to which a country can improve. Accordingly, the optimal solution $x$ * that we obtain here is exactly our optimal allocation. However, the biggest problem is that the objective function is not convex. Therefore, we cannot apply the classical tools for convex optimization to this problem. In order to solve this problem, our idea is to use a greedy algorithm. The way that we apply the greedy algorithm is that firstly we separate the total budget into k equal parts, and then for the first equal part we try to search for the product that can optimize our objective function by iterating through each $p \in P$. After we find the optimal product p for this equal part, we add the amount back to the export value of this product p and update the ECI for this country. If we observe that ECI does not change after iterating all the products, then we can add up one

more equal part to invest. Until we find there is an ECI increment, we add the amount now to that product. The equal rest parts can be done in the same way.

By definition, a greedy algorithm is a kind of local search algorithm that is possible to run into a local minimum. Therefore, for local search algorithms, one typical way to keep it away from the local minimum is to enhance the vision of the algorithm by looking more steps forward. In our setting, whenever we obtain an increment of ECI, we mark down the required amount of export growth. Simultaneously, we continue searching for the next ECI increment point, compare the ratios of these two possibilities and pick the largest one. The ratio is defined as the following:

$$Ratio(p) = \frac{ECI_p - ECI_0}{||x_p - x_0||_1}$$

where $ECI_p$ is just the ECI of the country after adding some amount of export increment to product $p$, which is exactly the amount that is able to make ECI change. Also, we can express it in the way of $||x_p - x_0||_1$.

In this way, we can obtain a further sighted greedy algorithm. Also, the number of parts that we would like to separate the budget into also matters. In order to solve this problem, we added a stabilizer on top of the enhanced greedy algorithm by running the algorithm for different numbers of parts and selecting the portfolio with the largest ECI increment and most complex economic structure.

Besides, it is quite common that there is a surplus in the budget. To deal with the surplus, we just add it to the product which needs the least amount of dollars to make $M_{cp} = 1$ (i.e., the smallest amount of required export increase).

**Economic Panel**

As stated above, the enhanced greedy algorithm is developed mainly for solving the optimal allocation strategy. In order to visualize this allocation, an economic panel is created, consisting of a pie chart, a line chart and a table. First, a pie chart is used to show the percentage of each product in the optimal portfolio. The number located in the center of the circle denotes the heuristic growth of total export, which is also our budget. Each category in the chart has a determined percentage of the allocated budget. The lines below the pie charts show how the ECI changes during the searching process of the greedy algorithm. Even though sometimes an increase of the export by 15% or 20% does not seem reasonable to developing countries, yet it can also give a sense of how a country should develop in the long term. Eventually, the graph in the end is just for numerical explanation for the pie charts.

**Interpretation**

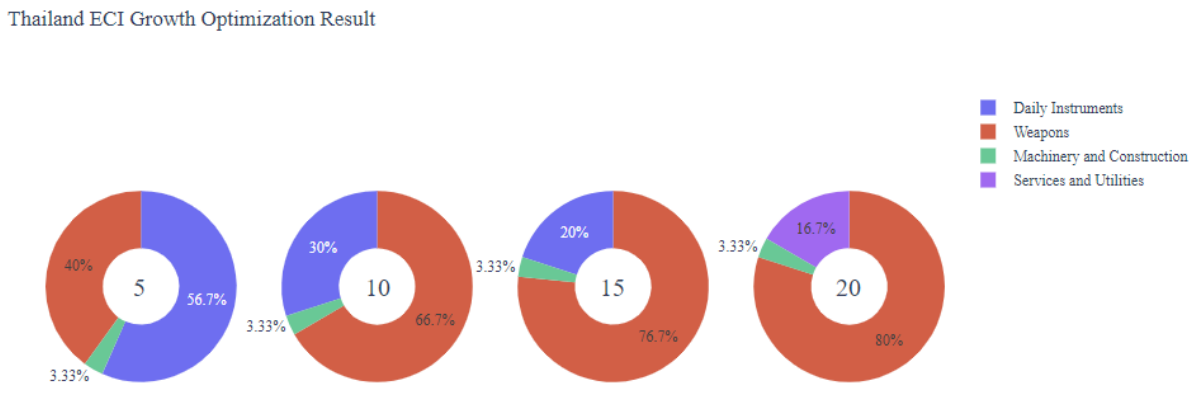
For the purpose of a clear explanation of the line charts, we label all the products in Table 7. Please refer to this table when trying to understand the investment trajectories in the line charts.

| Label | Product Names |
|---|---|
| 1 | Animal and By-products |
| 2 | Vegetables, Fruits and Plants |
| 3 | Food and Preparations |
| 4 | Energy Drilling and Mining |
| 5 | Chemicals |
| 6 | Chemical Manufacturing |
| 7 | Raw Manufacturing |
| 8 | Wood Products |
| 9 | Clothing and Accessories |
| 10 | Textile |
| 11 | Metals |
| 12 | Machinery and Construction |
| 13 | Transportation |
| 14 | Weapons |
| 15 | Daily Instruments |
| 16 | Precious Stones |
| 17 | High-complexity Products |
| 18 | Services and Utilities |

Table 7. Products Labeling

Figure 4 shows an economic panel for Thailand in 2023. We can see that for different percentages of heuristic total export growth, the ECIs of Thailand actually have different increments. From the Thailand ECI Increment plot, we can see that even a small export-growth target yields a substantial early gain in economic complexity. However, returns diminish as growth targets rise further. Specifically, 5% export growth lifts Thailand's ECI from a baseline of 1.20 to approximately 1.45, 10% growth pushes the ECI to 1.50, 15% growth plateaus at 1.50, and 20% growth brings Thailand's ECI to 1.60.

For 2023, a 5 percent increase in exports would emphasize Daily Instruments, followed by Weapons, Weapons, and Weapons for 10, 15, and 20 percent export growth, respectively. These strategies go in line with the suggested trajectories based on ECI. At 5% growth, half the portfolio lies in Daily Instruments, but as the growth target rises to 10%, 15%, and 20%, weapons steadily dominate, reaching 66.6%, 76.7%, and 80% of the portfolio, respectively. Meanwhile, Machinery and Construction stays constant at 3.3% of the portfolio, and Services and Utilities enters only at the 20% growth target. A more detailed numerical representation of the portfolio for Thailand is shown in Table 8.



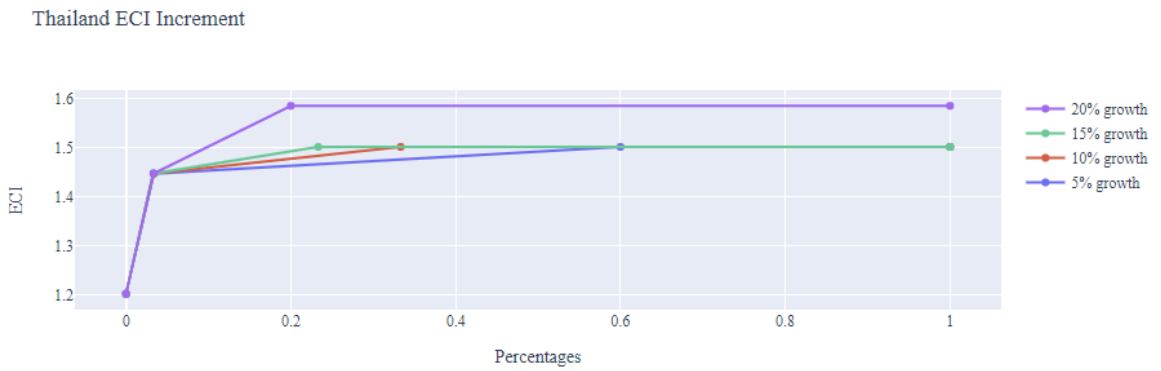



Figure 4. Greedy-Based ECI Improvement Results of Thailand with Different Budgets (2023)

| labels | 5% | 10% | 15% | 20% |
|---|---|---|---|---|
| Heuristic Export Growth | 5% | 10% | 15% | 20% |
| Machinery and Construction | 195806575.157000 | 391613150.314000 | 587419725.471000 | 783226300.627667 |
| Weapons | 2349678901.884000 | 7832263006.280001 | 13510653685.833000 | 18797431215.064003 |
| Daily Instruments | 3328711777.669000 | 3524518352.826000 | 3524518352.826000 | 0 |
| Services and Utilities | 0 | 0 | 0 | 3916131503.138334 |
| Increased ECI | 1.501761 | 1.501761 | 1.501761 | 1.585241 |

Table 8. Portfolio of Thailand (2023)

### 4.3 Sensitivity Analysis and Contingency Strategy

In the following section, our broad aim is to understand how sensitive Thailand is to changes in export in various industries. While the Economic Complexity Index has been shown to contain valuable information about the economic strength/capacity of a country, little has been said as to the stability of the measure itself. We seek to uncover the main drivers of stability under a country's ECI score and better understand longer-term projections for ECI growth. Additionally, gaining an intuition on the industries that are bolstering an economy and the industries that offer the greatest return per unit of investment is best done through visualization of industry-specific sensitivity. To this end, we have developed two categories of sensitivity analysis.

#### ECI Sensitivity Graph

First, in order to assess the effect particular industries have on each respective country, we developed ECI Sensitivity graphs (Figure 5). We use the shocking methodology to increment or decrement an industry's export level by a certain percentage and recalculate the ECI, all else being equal. While freezing the export dynamics of the rest of the world and enacting an instantaneous shock in export for a particular industry and country may not reflect real-life dynamics, what it does offer us is an understanding of a country's level of dependency on an industry in the context of ECI. The shocking is done in small percentage increments in order to smooth out the resulting curve. Additionally, a recurring theme one will see is that for most industries, a breaking point occurs where the ECI either jumps upwards or downwards, creating a hockey stick shape rather than a linear trend. While the inflection point may not be

accurate in the context of real life, comparing the various inflection points across industries can give us valuable insight.

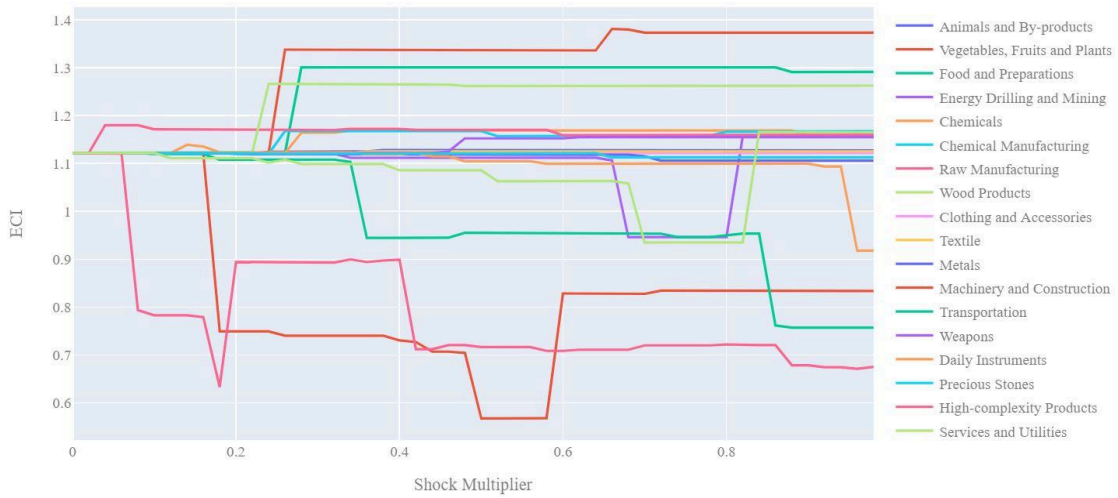



Figure 5. Sample ECI Sensitivity Graph (2018)

The graphs are formulated as follows. There are two ECI Sensitivity graphs for the year 2023. The first chart, Incremental ECI Sensitivity, shows positive shocking in which export numbers are shocked upwards per industry. The X-axis is in terms of percentage shock per current dollar value of export. The Y-axis is the recalculated ECI level post-shocking. The second chart, Decremental ECI Sensitivity, follows the same format except the X-axis represents negative shock. In other words, exports for each industry are decremented, and the ECI is accordingly recomputed.

### 4.3 Sensitivity Analysis Results and Interpretation

We examine Thailand's ECI Decremental and Incremental Sensitivity Charts for 2023. An increase in Raw Manufacturing will lead to the highest change in ECI, followed by Energy Drilling and Mining and Metals. The strong sensitivity effect of manipulating the Energy Drilling and Mining category changes from previous years. For the negative shock cases, Animals and By-products becomes most impactful, remaining consistent with the previous years. The rest of the industries do not appear to have the same level of sensitivity.

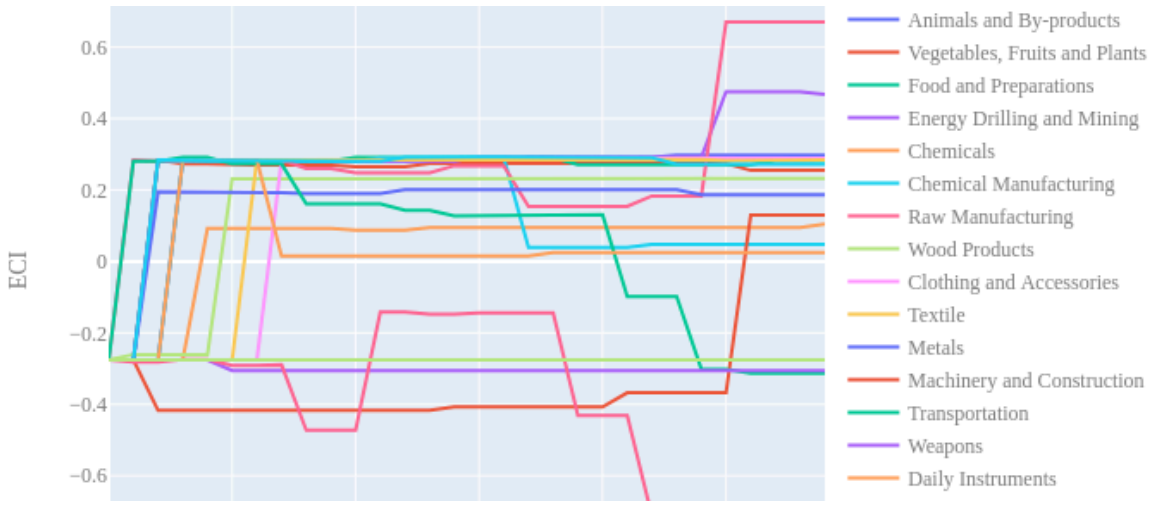



Figure 6a. Incremental ECI Sensitivity for Thailand (2023)

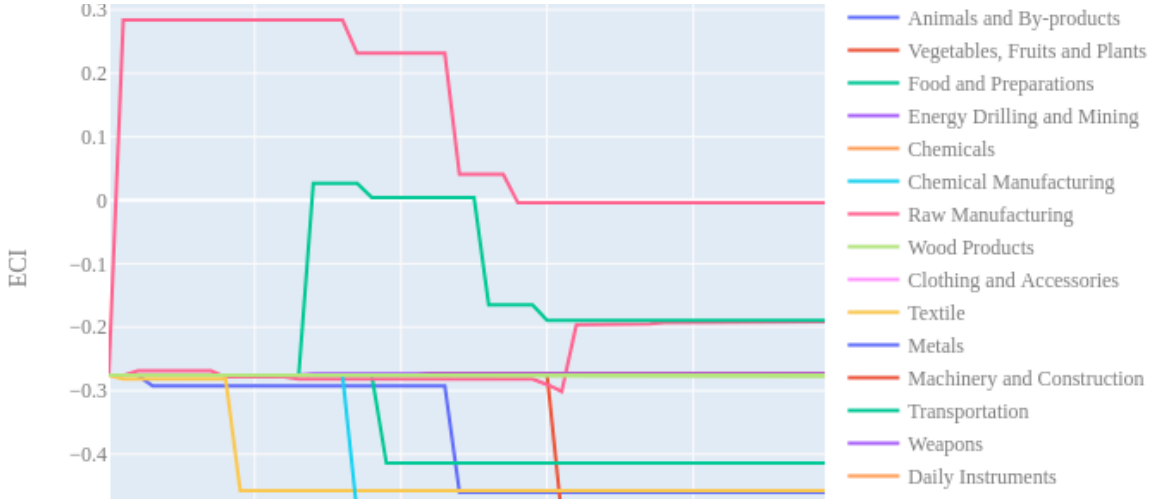



Figure 6b. Decremental ECI Sensitivity for Thailand (2023)

## 5. Macro-level Predictions

### 5.1 Structural Optimality Index

ECI sensitivity described in the section before has proven to be a significant factor in deciding the economic outcomes of countries. In order to quantitatively evaluate how effective our development strategies are and how deviations from the suggested developmental path will lead to suboptimal growth results, we define the measure – Structural Optimality Index (SOI) – on a country level as the Adjusted R-Square obtained from regressing the country's realized industry development over the suggested development strategy. Both strategies are quantified in nature, with the realized development represented by increasing exports in every major industry group and the suggested development strategy represented by the opportunity gains of developing the corresponding industry groups as calculated in previous sections. The adjusted R-square from the regression thus provides us with a quantified measure of how the development path adopted by the country has fitted with/differed from calculations on a complexity level.

After obtaining this measure for all countries in our database, we further regress countries' five-year economic growth, which is defined as the percent increase in GDP (USD), over

their SOI. Figure 9 illustrates the regression output, which shows a strong positive correlation between countries' SOI and their five-year GDP growth, significant at the 23.5% level and achieving an adjusted $R^2$ of 0.4%. In other words, the more a country develops, as suggested by calculations on a complexity level, the better economic outcomes it achieves.

| **Table 14: OLS Regression Results** | |
|---|---|
| **Dep. Variable:** GDP | **F-Statistic:** 1.422787384846241 |
| **$R^2$:** 0.012326746018551216 | **Prob (F-Stat):** 0.23542189042438527 |
| **Adj $R^2$:** 0.003662945545029661 | **No. Observations:** 116 |

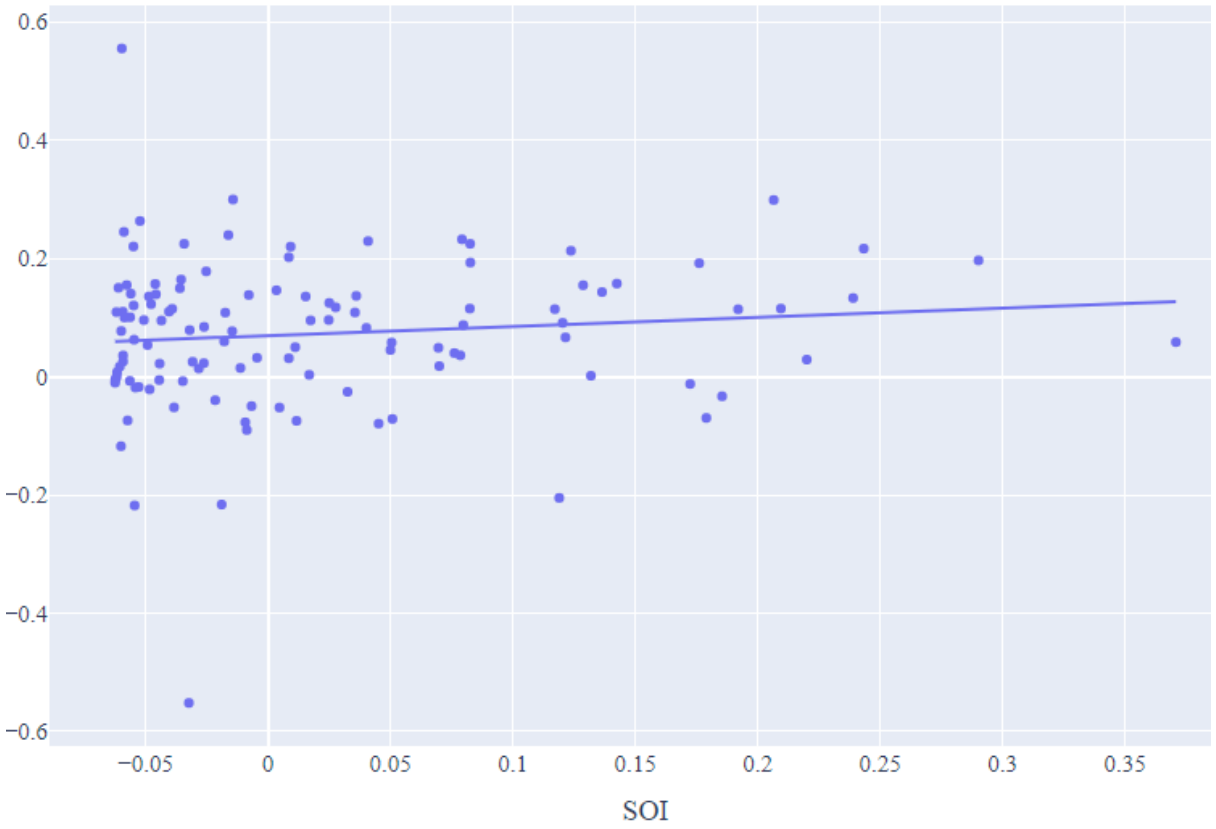



Figure 9. Regression of Five-Year GDP Growth over SOI (2019-2023)

### 5.2 SOI of Thailand and Its Implications

During the first examined period, Thailand, for instance, has a low level of structural optimality as compared with its developing peers. As the regression in Figure 10 suggests (the red curve represents export increase and the blue curve stands for opportunity gain of the corresponding industry), Thailand demonstrates a moderate level of correlation with the suggested developmental path, having an adjusted R-square of around 0.179.

Thailand ranks in the third quartile among all countries in terms of SOI and has made a slow rate of economic progress among its emerging peers from 2019 to 2023 with a 5-year GDP growth rate of -1.0%, which is consistent with the regression result in Figure 4. This is in direct contrast to emerging economies such as Guyana, where there is a high SOI of around

35% and corresponding GDP growth of approximately 8.2% from 2019 to 2023, making it one of the fastest developing economies during the period.

Specifically, Thailand's economic regression from 2019 to 2023 can be at least partially explained by a development strategy that demonstrates a low level of matching with our suggested developmental path. Specifically, Thailand invested heavily in the High-complexity Products industry, while it should have focused more on industry groups such as Machinery and Construction.

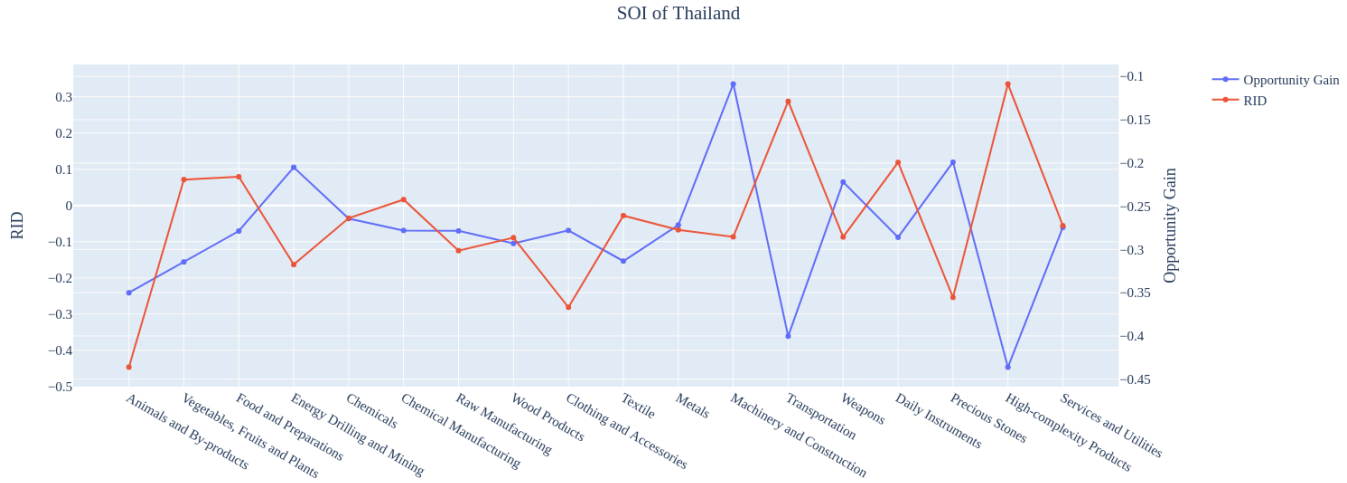



Figure 10. SOI of Thailand (2019-2023)

## 6. Concluding Remarks

In conclusion, Thailand has demonstrated a low level of economic growth in terms of annual GDP increase during the period from 2013 to 2023, concurrent with an decrease in economic complexity as compared with major developing economies. For 2019-2023, the growth in GDP has decreased alongside a large decrease in ECI, in contrast with its comparable countries, and its ECI became much lower than that of some of the leading economies in the world.

We believe that the consistent progress during the first period can be partially explained through the analysis of the economic and product complexity structure of Thailand on the industry group level. The product classification system we adopt allows us to measure the realized industry development for various industry groups and to observe that Thailand progressed in developing most of its industries during the period, while its developing counterparts and emerging economies around the globe adopted much more complex and progressive paths.

Based on our analysis of the opportunity gain and sensitivity analysis of major industry groups for 2013-2018, Thailand could have furthered its gains by prioritizing Services and Utilities and Wood Products. We further note that at the time, Thailand had a slightly mismatched development strategy that indicated a moderate level (compared to other developing countries) of structural optimality on the complexity level, having an adjusted R-square of around -5.8%. Since a significant positive correlation exists between five-year GDP growth and the structural optimality index of an economy as we define and calculate, Thailand indeed had a suboptimal growth experience as compared with fast-growing economies both on a regional scale and on a global scale and had the potential to further its economic gains through industrial structure optimization as suggested in the paper.

During the second examined period from 2019 to 2023, Thailand performed much weaker than before and compared to other countries, particularly Chile. We infer that during the COVID-19 period, Thailand focused on developing higher complexity industries despite a lower demand for these industries given the global economic downturn. We also note that Thailand invested heavily in the High-complexity Products and Transportation industries, whereas it should have focus more on industry groups such as Machinery and Construction and Energy Drilling and Mining, which leaves some room for strategic improvement. Nonetheless, Thailand invested optimally in Wood Products and Services and Utilities, which contribute the most to its advancements. The ECI suggests that greater growth could have been achieved through Raw Manufacturing, followed by Energy Drilling and Mining and Metals. The surprising growth also increased the adjusted R-squared value to 17.9% in 2023 from -5.8% in 2018, demonstrating that the pursued strategies contributed to the GDP. With the revised focus, Thailand has further potential to advance its economic growth to recover its position among other countries to pursue its post-pandemic recovery.

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